EDGE COMPUTING DRIVEN SMART GRID OPTIMIZATION FOR RESIDENTIAL ENERGY MANAGEMENT WITH ML

Final Phase Presentation on Major Project of the AY: 2024-25

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- Hardware and Software Specification
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- Feature Extraction
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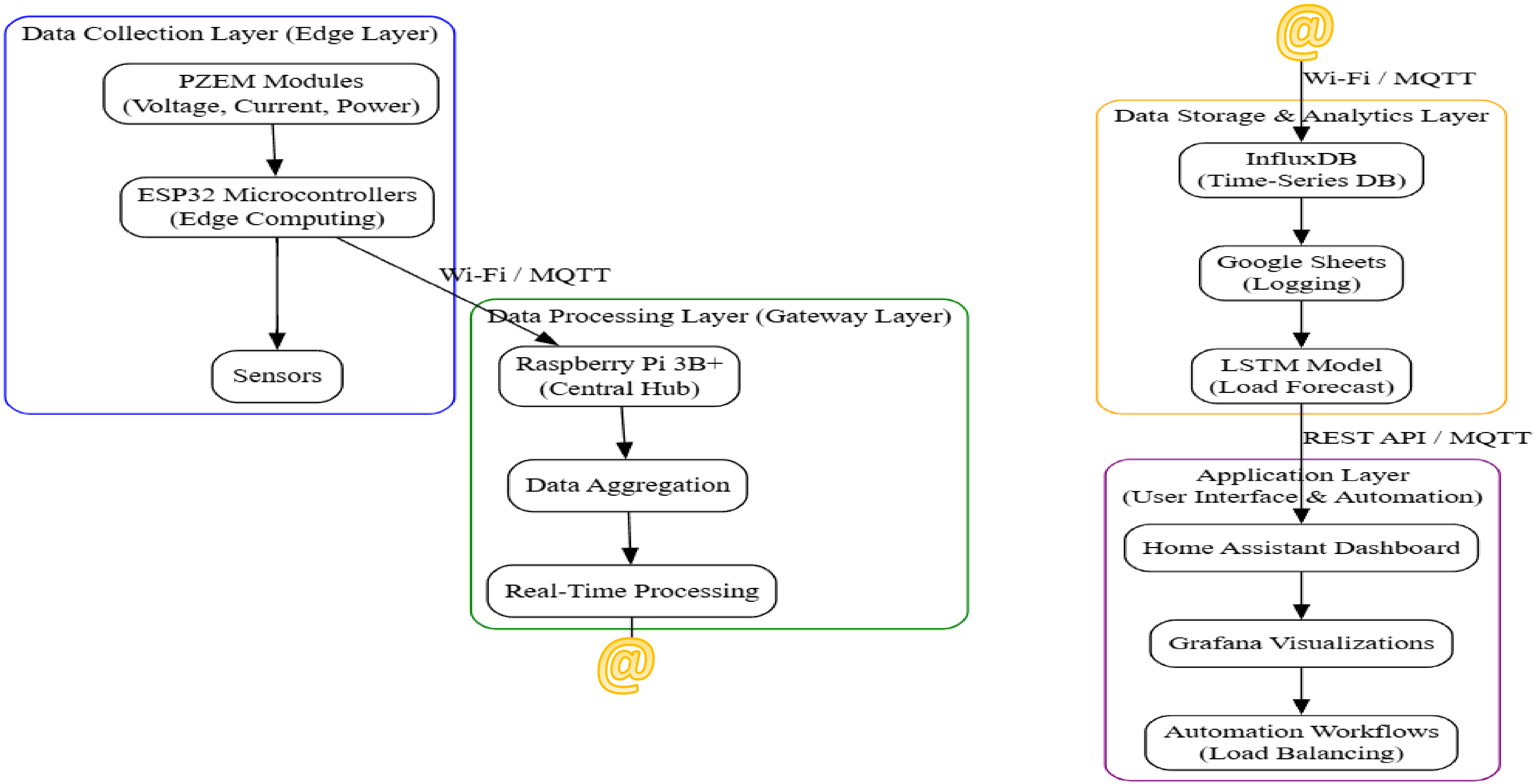
Objectives

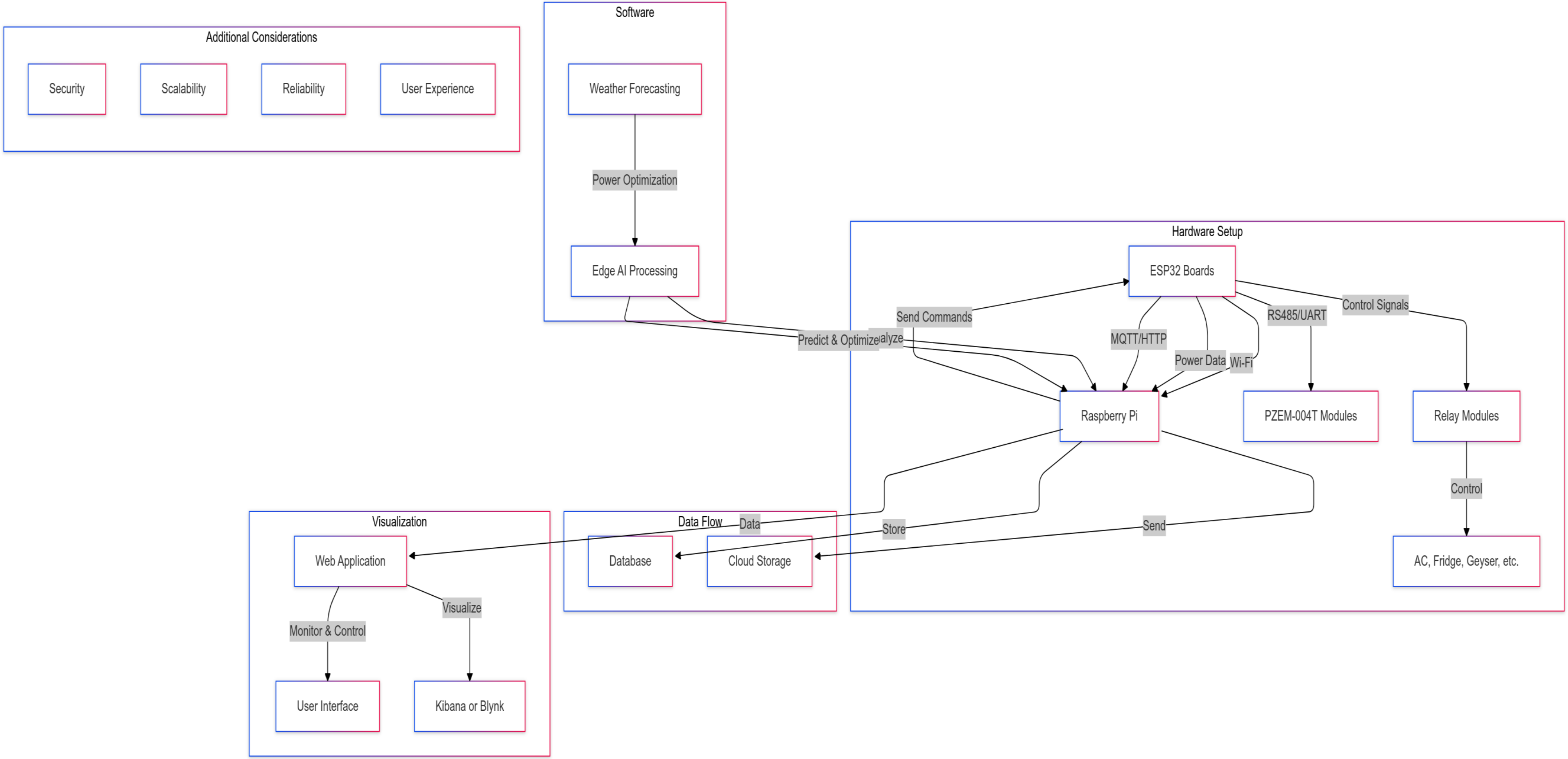
The following are the proposed objectives of the project based on the research gaps:

- To enable real-time decision-making in residential power grids using Raspberry Pi as edge devices.
- To enable efficient integration and management of solar energy in residential power grids using edge computing to monitor and control solar generation and storage systems.
- To enhance energy management efficiency in smart grids using data-driven analytics, and reduce latency by utilizing edge resources near data sources for improved residential power efficiency.
- To improve reliability and resilience of residential power grids through optimized distributed processing.
- To develop a cost-efficient system for an efficient, reliable, and scalable residential smart grid.

Proposed Methodology



Proposed Methodology



Hardware Specification

- **PZEM Modules**
  - To measure voltage, current, power, and energy consumption of individual appliances
  - Interface PZEM modules with ESP32 microcontrollers

- **ESP32 Microcontrollers**
  - To collect and preprocess data from PZEM modules and transmit it to the Raspberry Pi
  - Flash ESP32 devices with ESPHome firmware for seamless integration with Home Assistant
  - Configure Wi-Fi settings to enable wireless communication with the Raspberry Pi

Hardware Specification

- **Raspberry Pi 3B+**
  - Acts as the central hub for data aggregation, processing, and storage.
  - Install the Raspberry Pi OS on a microSD card.
  - Connect the Raspberry Pi to a monitor, keyboard, and mouse for initial setup.
  - Connect the Raspberry Pi to the same Wi-Fi network as the ESP32 devices.
  - Use a 5V power adapter to power the Raspberry Pi.
  - Connect to the internet via Wi-Fi or Ethernet.

Software Specification

- **Home Assistant**
  - To provide a user-friendly interface for real-time monitoring and automation.
  - Install Home Assistant on the Raspberry Pi using the official installation guide.
  - Configure the ESP32 devices in Home Assistant using ESPHome.
- **Grafana**
  - For creating advanced visualizations for energy consumption and forecasts.
  - Install Grafana on the Raspberry Pi or a cloud server.
  - Create dashboards to display real-time energy consumption, historical trends, and load forecasts.

Software Specification

- **LSTM Model**

- To forecast future energy consumption based on historical data, and balance load
  - Collect historical energy consumption data from PZEM modules
  - Preprocess the data (e.g., normalize, handle missing values)
  - Train the LSTM model using a deep learning framework, TensorFlow

Dataset Description

The dataset has **47 features**. The features are categorized into:

1. General Information

- **Day**: The day of the data recording (integer)
- **Weather**: Weather conditions (e.g., Sunny, Rainy, Cloudy)
- **Temperature (°C)**: Ambient temperature in °C
- **Voltage (V)**: Voltage supply to the household

2. Electrical Appliances Usage

- **Hours Used**: The number of hours the appliance was used
- **Energy Consumption (kWh)**: The energy consumed in kWh
- **Current (A)**: The current drawn by the appliance in amperes

Dataset Description

2. Electrical Appliances Usage

The monitored appliances include:

- **Air Conditioners**: Bedroom and Master Bedroom
- **Fans**: Living Hall and Master Bedroom
- **Lights**: Bedroom, Living Hall, Kitchen, and Master Bedroom
- **Microwave**
- **Geysers**: Two separate geysers
- **Washing Machines**: Two washing machines
- **Fridge**

3. Total Energy Consumption (kWh)

- The sum of energy used by all appliances in kWh

Feature Extraction

- Feature extraction is the process of selecting raw data into meaningful features to improve model performance
- It helps reduce dimensionality, enhance predictive accuracy, and make data more interpretable.

```
Sample Data:
   Day Weather  Temperature (°C)  Voltage (V)  AC Shishir Room Hours Used  \
0    1   Sunny              21.6       223.38                        5.07
1    2   Rainy              37.5       221.22                        4.09
2    3  Cloudy              36.9       223.84                        4.38
3    4   Sunny              38.6       224.04                        5.02
4    5  Cloudy              23.3       223.72                        4.54

   AC Shishir Room Energy (kWh)  AC Shishir Room Current (A)  \
0                         4.637                         4.09
1                         3.744                         4.13
2                         4.009                         4.09
3                         4.590                         4.08
4                         4.151                         4.09
```

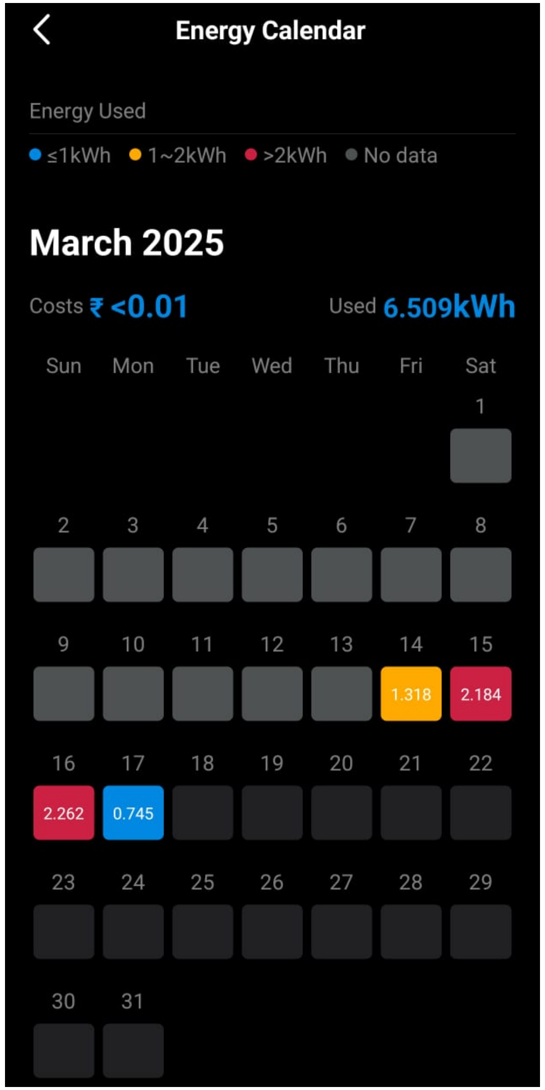
Modelling Techniques

- **Long Short-Term Memory (LSTM)** networks are employed for load forecasting and a rule-based allocation strategy for dynamic load balancing
- LSTM is a type of **Recurrent Neural Network (RNN)** designed to handle sequential data
- Used to predict **Total Energy Consumption (kWh)** based on historical patterns and external factors
- Key steps in LSTM modeling:
  - Data preprocessing: Feature scaling and reshaping into time-series format
  - Model architecture: 2 LSTM layers with dropout and dense layers for final prediction
  - Training with **Adam optimizer and early stopping** to prevent overfitting

Hardware Set Up

Experimental Results

- The image shows Home Assistant interface, an energy monitoring dashboard that tracks electricity consumption in real time.
- It shows that the total consumption for the month so far is 6.509 kWh.



Experimental Results

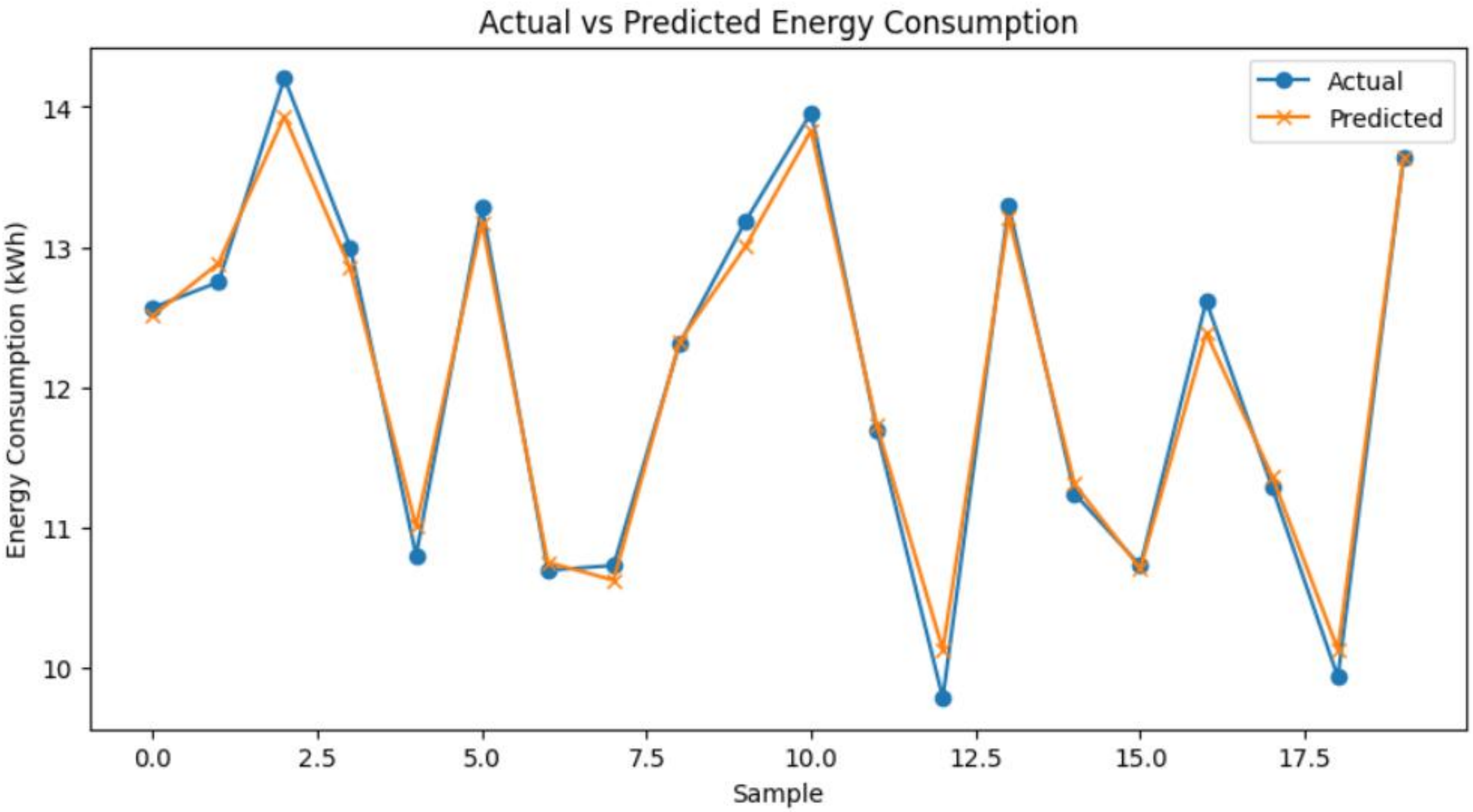


Fig.: Load Forecasting Actual vs Predicted Energy Consumption (Jupyter Notebook)

Experimental Results

- The MSE value of **0.0259** indicates a low average squared difference between predicted and actual values, suggesting minimal error.

- The RMSE of **0.1609** further confirms the model's accuracy, as lower values imply better performance.

- The R^2 value of **0.9852** demonstrates a strong correlation between the predictions and actual values.

- These metrics collectively suggest that the model is highly accurate and well-suited for predictive tasks.

Metric	Value
MAE	0.1340
MSE	0.0259
RMSE	0.1609
R^2	0.9852

Table: Accuracy of LSTM Model

Experimental Results

- For each time interval, the predicted demand is matched entirely by the grid.

- No other energy sources, such as solar or wind, were utilized. There is no unfulfilled demand.

- The energy system is effectively supplying the required power without any shortages.

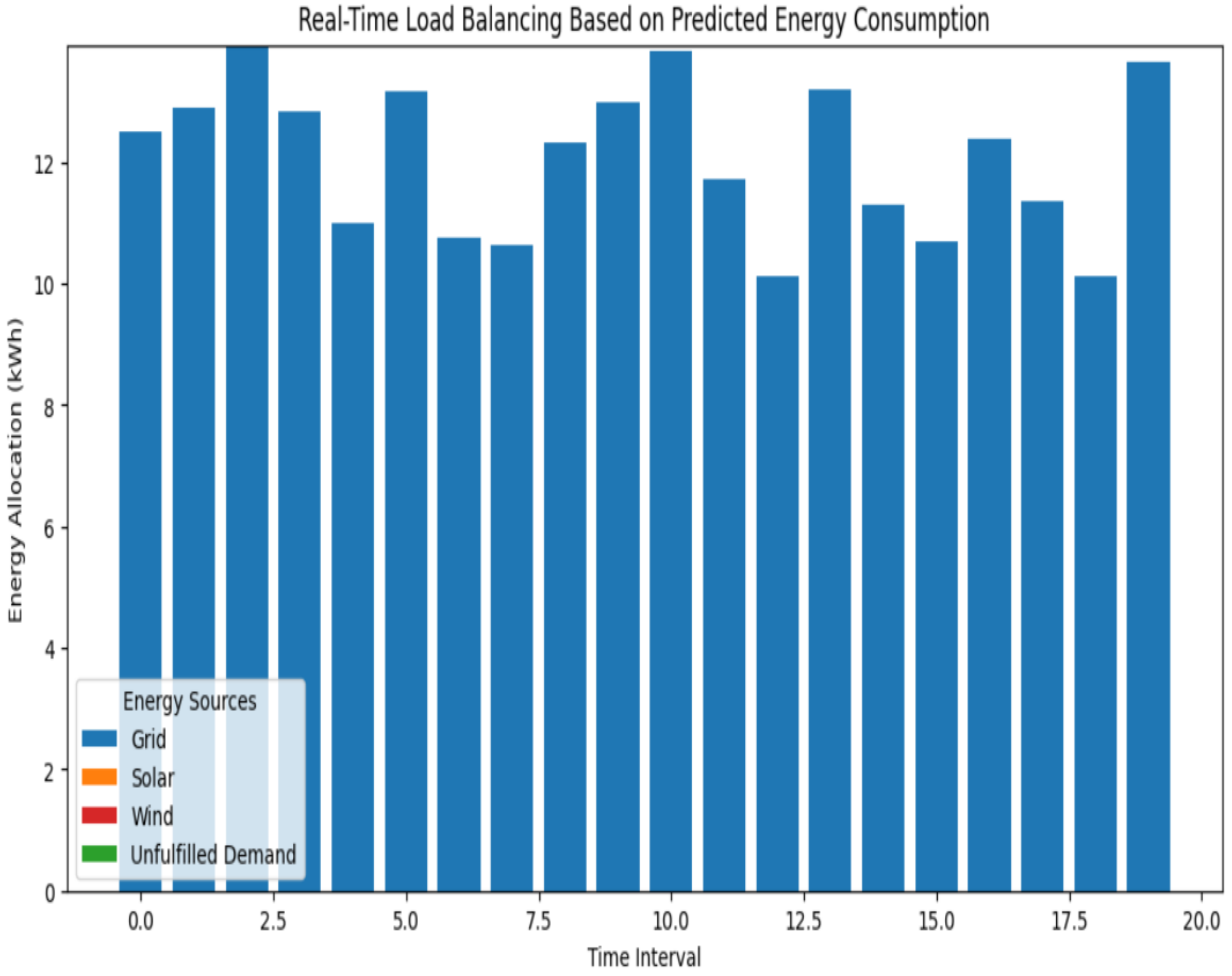


Fig.: Load Balancing Based on Predicted Energy Consumption

Experimental Results

Fig.: Electricity Bill Before and After Optimization

Analysis of Results

- The **energy monitoring dashboard** provides insights into daily and monthly energy consumption trends, showing variations in energy usage over time.
- The **actual vs. predicted energy consumption** graph shows a high correlation between predicted and actual values, demonstrating the accuracy of the predictive model **(R^2=0.98)**.

Metric	Cloud Model	Edge Model
Latency	~250ms–1s	~50ms
Energy Efficiency	Baseline	+48%
Peak Load Reduction	Poor	56%
Cost Savings	Minimal	15%

Analysis of Results

- The **performance metrics** confirm that the model effectively forecasts energy demand with minimal error.
- The system achieves an **80% reduction in latency**, with forecasting accuracy yielding an **MAE** of **0.13** and **RMSE** of **0.16** under test conditions.
- Dynamic load balancing leads to a **48% improvement in energy efficiency** and a **56% reduction in peak load**, while the complete system implementation yields a **15% reduction in energy costs**.

Future Scope

- **Renewable Energy Integration**
  - Incorporate solar and wind energy for optimized load balancing and reduced grid dependency.
- **System Integration**
  - Use **Raspberry Pi** as a centralized platform to integrate sensors, forecasting models, and load balancing mechanisms.
  - Utilize **Raspberry Pi** for local processing, ensuring an affordable and efficient solution.

Future Scope

- **Scalability and Performance**
  - Expand **distributed edge computing** to support larger residential areas.
  - Reduce computational overhead and enhance system performance.
- **Real-Time Analytics & Monitoring**
  - Strengthen cloud integration for **remote energy monitoring** and predictive maintenance.

Conclusion

- Implemented **edge computing-driven smart grid optimization** to enhance residential energy management.
- Achieved **real-time load forecasting** and **efficient load balancing** using LSTM-based predictions.
- Results demonstrate **high prediction accuracy** with minimal error and strong correlation between actual and predicted values.
- The system ensures **stable energy distribution**, reducing reliance on the grid.
- Future enhancements include **renewable energy integration,** and **improved scalability.**